Syllabus

Course Information

Course Number:	*CSCE 636*
Course Title:	*Deep Learning*
Section:	*700*
Time:	*Mondays and Wednesdays (online)*
Location:	*online*
Credit Hours:	*3*

Instructor Details

Instructor:	*Anxiao Jiang*
Office:	*425 PETR*
Phone:	*979-845-7983*
E-Mail:	*ajiang@cse.tamu.edu*
Office Hours:	*9:00-9:30am on Mondays via zoom*

Course Description

Foundations and advanced methods in deep learning, including principles for designing, training, and evaluating deep neural networks. Topics include supervised learning, loss functions, back-propagation, regularization, and modern optimization techniques; architectures for structured and unstructured data such as convolutional networks for images and transformer models for sequences. Covers advanced topics including self-supervised learning, representation learning, deep generative models. Emphasis on both theoretical understanding and practical implementation, with applications to vision and language tasks.

Course Prerequisites

Graduate classification or approval of instructor

Special Course Designation

N/A

Course Learning Outcomes

Upon finishing this class, students should be able to:

- Develop deep learning models for various classification, regression and generation tasks
- Design algorithms and software for training deep learning models
- Analyze the performance of deep learning models

- Design hyper-parameter tuning and regularization techniques for optimizing models

Textbook and/or Resource Materials

Required Textbook

Deep Learning with Python, Third Edition, By Francois Chollet and Matthew Watson, Manning Publications, November 2025

Grading Policy

Six homework assignments: 25%
Three projects: 25% each (75% total)

Grading Scale: A = [90, 100], B = [80, 90), C = [70, 80), D = [60, 70), F = [0, 60)

Late Work Policy

With the exception of excused absences (see Student Rule 7), no homework or project submitted after the established deadline will be accepted.

Course Schedule

Week 1
Lecture Topics: Mathematical Building Blocks of Deep Learning
Reading List: Chapter 1-2
Notes and Assignment:

Week 2
Lecture Topics: Programming Languages for Deep Learning
Reading List: Chapter 3
Notes and Assignment:

Week 3
Lecture Topics: Deep Learning for Classification and Regression
Reading List: Chapter 4
Notes and Assignment:

Week 4
Lecture Topics: Fundamental Machine Learning Techniques for Deep Learning, Universal Workflow of Deep Learning
Reading List: Chapter 5-6
Notes and Assignment:

Week 5
Lecture Topics: Dive Deep into Deep Learning Programming
Reading List: Chapter 7
Notes and Assignment:

Week 6
Lecture Topics: Convolutional Neural Networks
Reading List: Chapter 8
Notes and Assignment:

Week 7
Lecture Topics: Fundamental Deep Learning for Computer Vision
Reading List: Chapter 8
Notes and Assignment:

Week 8
Lecture Topics: Advanced Deep Learning for Computer Vision
Reading List: Chapter 9
Notes and Assignment:

Week 9
Lecture Topics: Deep Learning for Time Series; LSTM and GRU Models
Reading List: Chapter 10
Notes and Assignment: Project 1 due on 3/16/2026

Week 10
Lecture Topics: Deep Learning for Text; Self-Attention Mechanism and Transformer Models
Reading List: Chapter 11
Notes and Assignment:

Week 11
Lecture Topics: Generative Deep Learning
Reading List: Chapter 12
Notes and Assignment:

Week 12
Lecture Topics: Auto-Encoder and Generative-Adversarial Network Models
Reading List: Chapter 12
Notes and Assignment: Project 2 due on 4/6/2026

Week 13
Lecture Topics: Hyper-Parameter Optimization for Deep Learning
Reading List: Chapter 13
Notes and Assignment:

Week 14

Lecture Topics: Deep Reinforcement Learning
Reading List:
Notes and Assignment: Project 3 due on 4/24/2026

Week 15
Lecture Topics: Deep Reinforcement Learning
Reading List:
Notes and Assignment:

Optional Course Information Items

Consider adding the following additional information items to the course syllabus when appropriate. Delete any information and/or subheadings if not needed, including this note.

Technology Support – Provide appropriate technical support information to inform students who to contact if they encounter technical difficulties (e.g., direct technical questions to the course teaching assistant; contact the vendor; etc.). Technical support information should include information such as who to contact, how to contact that resource, hours of availability, etc.

Texas A&M at Qatar
Texas A&M University at Qatar students can also direct their technical questions to servicedesk@qatar.tamu.edu

Learning Resources – Provide information regarding available learning resources such as supplemental instruction or tutoring when appropriate (e.g., information about the University Writing Center for a W/C designated course or related LinkedIn Learning modules appropriate for the course topic). The Study Hub website lists many on-campus learning resources to support students in achieving academic excellence.

Texas A&M at Galveston
On-campus learning resources to support students in achieving excellence are available through The Center for Academic Learning Support (tamug.edu/cals).

Texas A&M at Qatar
Texas A&M University at Qatar students should contact the Center for Teaching and Learning at ctl@qatar.tamu.edu for questions related to learning support, peer tutoring, supplemental instruction, writing support, etc.

University Policies

This section outlines the university-level policies that must be included in each course syllabus. The TAMU Faculty Senate established the wording of these policies.

***NOTE**: Faculty members should not change the written statements. A faculty member may add separate paragraphs if additional information is needed.*

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to Student Rule 7 in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to Student Rule 7 in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" (Student Rule 7, Section 7.4.1).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" (Student Rule 7, Section 7.4.2).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See Student Rule 24.)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" (Section 20.1.2.3, Student Rule 20).

Texas A&M at College Station
You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Texas A&M at Galveston
You can learn more about the Honor Council Rules and Procedures as well as your rights and responsibilities at tamug.edu/HonorSystem.

Texas A&M at Qatar

You can learn more about academic integrity and your rights and responsibilities at Texas A&M University at Qatar by visiting the Aggie Honor System website.

Notice of Nondiscrimination

Texas A&M University is committed to providing safe and non-discriminatory learning, living, and work environments for all members of the University community. The University provides equal opportunity to all employees, students, applicants for employment or admission, and the public regardless of race, color, sex (including pregnancy and related conditions), religion, national origin, age, disability, genetic information, or veteran status. Texas A&M University will promptly, thoroughly, and fairly investigate and resolve all complaints of discrimination, harassment (including sexual harassment), complicity and related retaliation based on a protected class in accordance with System Regulation 08.01.01, University Rule 08.01.01.M1, Standard Administrative Procedure (SAP) 08.01.01.M1.01, and applicable federal and state laws. In accordance with Title IX and its implementing regulations, Texas A&M does not discriminate on the basis of sex in any educational program or activity, including admissions and employment. The following person has been designated to handle inquiries and complaints regarding the non-discrimination policies: Jennifer M. Smith, TAMU Associate VP & Title IX Coordinator at YMCA Ste 108, College Station, TX 77843, 979-458-8407, or email civilrights@tamu.edu. For other reporting options, visit https://ocrcas.ed.gov/contact-ocr to locate the address and phone number of the office that serves your area, or call 1-800-421-3481.

Civil Rights, Free Speech, and Title IX Policies

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit discrimination and harassment based on an individual's race, color, sex, (including pregnancy and related conditions), religion, national origin, age, disability, genetic information, veteran status, or any other legally protected characteristic. This includes forms of sex-based violence, such as sexual assault, sexual harassment, sexual exploitation, dating/domestic violence, and stalking.

Students can report discrimination/harassment, access supportive resources, or learn more about their options for resolving complaints on the University's Civil Rights & Title IX webpage.

Students should be aware that all university employees (except medical or mental health providers) are mandatory reporters, which means that if they observe, experience or become aware of an incident that they reasonably believe to be discrimination/harassment alleged to have been committed by or against a person who was a student or employee at the time of the incident, the employee must report the incident to the university.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below) Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or

chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.
To request academic accommodations, contact the designated ADA office based on your location:

- Texas A&M University, College of Nursing, College of Dentistry, Irma Lerma Rangel College of Pharmacy College Station, College of Medicine, School of Public Health, Institute of Biosciences and Technology, EnMed Program, Bush School in Washington DC, Mays Business School – CityCentre, TAMU Engineering Academies, Texas A&M University Higher Education Center at McAllen and Texas A&M University at Galveston should contact Disability Resources at (979) 845-1637 or disability@tamu.edu.
- Texas A&M University School of Law should contact the Office of Student Affairs at (817) 212-4111 or law-disability@law.tamu.edu to request accommodations.
- Irma Lerma Rangel College of Pharmacy in Kingsville should contact the Disability Resource Center at Texas A&M University - Kingsville at (361) 593-3024 or drc.center@tamuk.edu to request accommodations.
- Texas A&M University College of Veterinary Medicine & Biomedical Sciences in Canyon should contact the Office of Student Accessibility at West Texas A&M University – Canyon at (806) 651-2335 or osa@wtamu.edu.
- Texas A&M University at Qatar (TAMUQ) should contact the campus psychologist, Dr. Steve Wilson +974-4423-0047 or stephen.wilson@qatar.tamu.edu.

If you are experiencing difficulties with your approved accommodations, contact the office responsible for approving your accommodations or the Texas A&M ADA Coordinator Julie Kuder at ADA.Coordinator@tamu.edu or (979) 458-8407.

Pregnancy Accommodations

Texas A&M provides reasonable accommodations to students due to pregnancy and/or related conditions, such as childbirth, recovery and lactation. Students should contact the University's Pregnancy Coordinator as soon as they become aware of the need for accommodation. Depending on the circumstances, accommodations could include extended time to complete assignments or exams, changes in course sequence, or modifications to the physical classroom environment. Texas A&M will also allow a voluntary leave of absence, ensure the availability of lactation space, and maintain grievance procedures to provide for the prompt and equitable resolution of complaints of sex discrimination. For information regarding pregnancy accommodations, email TIX.Pregnancy@tamu.edu.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through University Health Services on its mental health webpage. The TELUS Health Student Support app provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or 988lifeline.org.

Texas A&M College Station

Students needing a listening ear can contact University Health Services (979.458.4584) 24-hour emergency help is also available through the 988 Suicide & Crisis Lifeline (988) or at 988lifeline.org.

Texas A&M at Galveston
Students who need someone to talk to can call (409) 740-4736 from 8:00 a.m. to 5:00 p.m. weekdays or visit tamug.edu/counsel for more information. For 24-hour emergency assistance during nights and weekends, contact the TAMUG Police Dept at (409) 740-4545. 24-hour emergency help is also available through the 988 Suicide & Crisis Lifeline (988) or at 988lifeline.org.

Texas A&M at Qatar
Texas A&M University at Qatar students wishing to discuss concerns in a confidential setting are encouraged to visit the Health and Wellness website for more information.

Statement on the Family Educational Rights and Privacy Act (FERPA)

FERPA is a federal law designed to protect the privacy of educational records by limiting access to these records, to establish the right of students to inspect and review their educational records and to provide guidelines for the correction of inaccurate and misleading data through informal and formal hearings. Currently enrolled students wishing to withhold any or all directory information items can do so within howdy.tamu.edu using the Directory Information Withholding Form. The complete FERPA Notice to Students and the student records policy is available on the Office of the Registrar webpage.
Items that can never be identified as public information are a student's social security number, citizenship, gender, grades, GPR or class schedule. All efforts will be made in this class to protect your privacy and to ensure confidential treatment of information associated with or generated by your participation in the class.

Directory items include name, UIN, local address, permanent address, email address, local telephone number, permanent telephone number, dates of attendance, program of study (college, major, campus), classification, previous institutions attended, degrees honors and awards received, participation in officially recognized activities and sports, medical residence location and medical residence specialization.

Campus-Specific Policies

Campuses may establish their own policies and minimum syllabus requirements. As long as these policies and requirements do not contradict the university level requirements, Campuses can add them in this section. Please remove this section if not needed.

College and Department Policies

College and departmental units may establish their own policies and minimum syllabus requirements. As long as these policies and requirements do not contradict the university level requirements, colleges and departments can add them in this section. Please remove this section if not needed.

Optional Syllabus Statements

The following statements below are optional. Leave as is to include, or delete if preferred. Either way, delete this note.

Free Speech and Civil Discourse

Texas A&M recognizes that the pursuit of truth through open and robust discourse is critical to academic inquiry. However, as a community of scholars, the university has an aspirational expectation that such discourse will be conducted in accordance with Aggie Core Values. In this "marketplace of ideas," we encourage civil dialogue creating an environment that allows individuals to express their ideas and to have their ideas challenged in respectful and responsible ways. Students can learn more about Freedom of Expression and Free Speech on the University's website about the First Amendment.

Artificial Intelligence Statements

There are two recommended syllabus statement options for instructions considering adding a statement about the use of Artificial Intelligence in their course. These are located on the Teach with AI webpage hosted at ai.tamu.edu. Faculty are strongly encouraged to have a written policy on AI use in their course to avoid any confusion regarding what is or is not permitted.